Building the Synoptic Key of Life website

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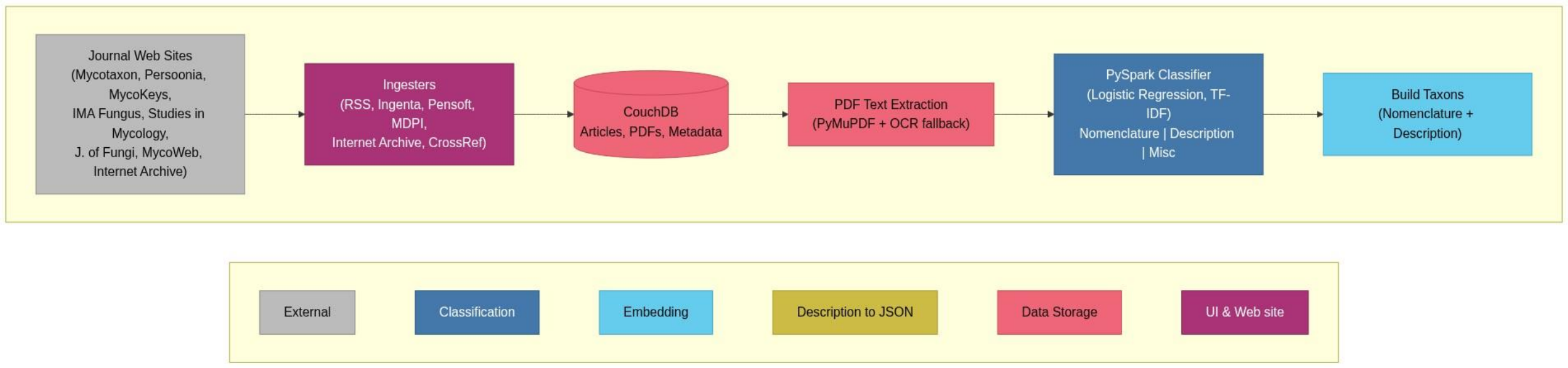
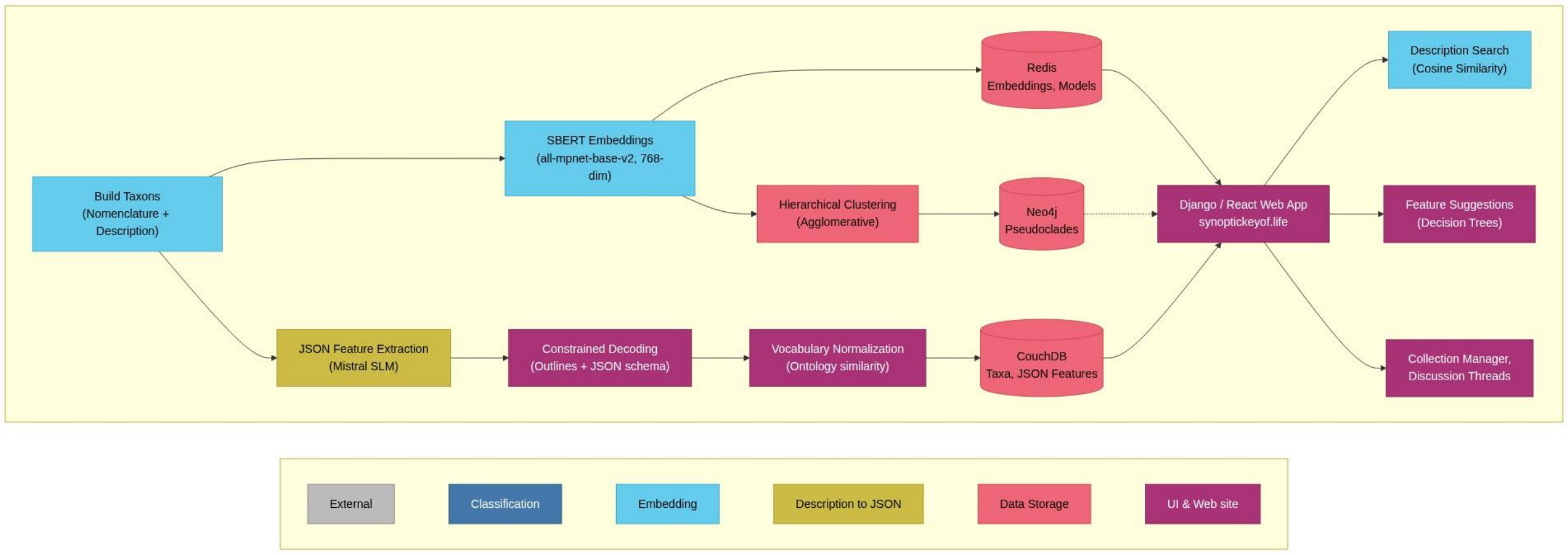
- MASMC 2026
- 2026-04-25

- North Carolina Agricultural and Technical State (NCA&T) University

SKOL Architecture by Implementation Project (1 of 2)



SKOL Architecture by Implementation Project (2 of 2)



YEDDA Annotation (Classification)

[@2. Vibrissea filisporia (Bonord.) Korf & A. Sánchez f. boudieri A. Sánchez & Korf in Sánchez, J. Agric. Univ. Puerto Rico 51: 83, 1967.
FIGS. 3, 4 & 5.#Nomenclature*]

[@APOTHECIA sessile, discoid, 0.75-1 mm in diameter. RECEPTACLE brown. DISC convex, buff when rehydrated. HYMENIUM making up to 3/4 of the apothecium, asci taking up most of the height. ECTAL EXCIPULUM brown, 65.3-76.4 µm wide, of textura globulosa, with cells turning outwards#Description*]

Classification

Dr Draft's state-of-the-art (SOTA) Literature Search

SBERT.net

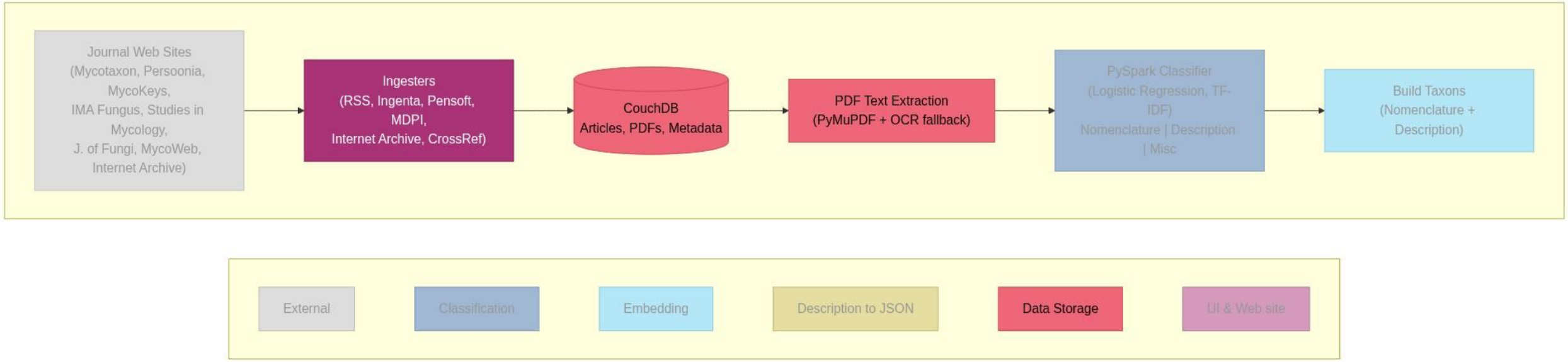
Embedding

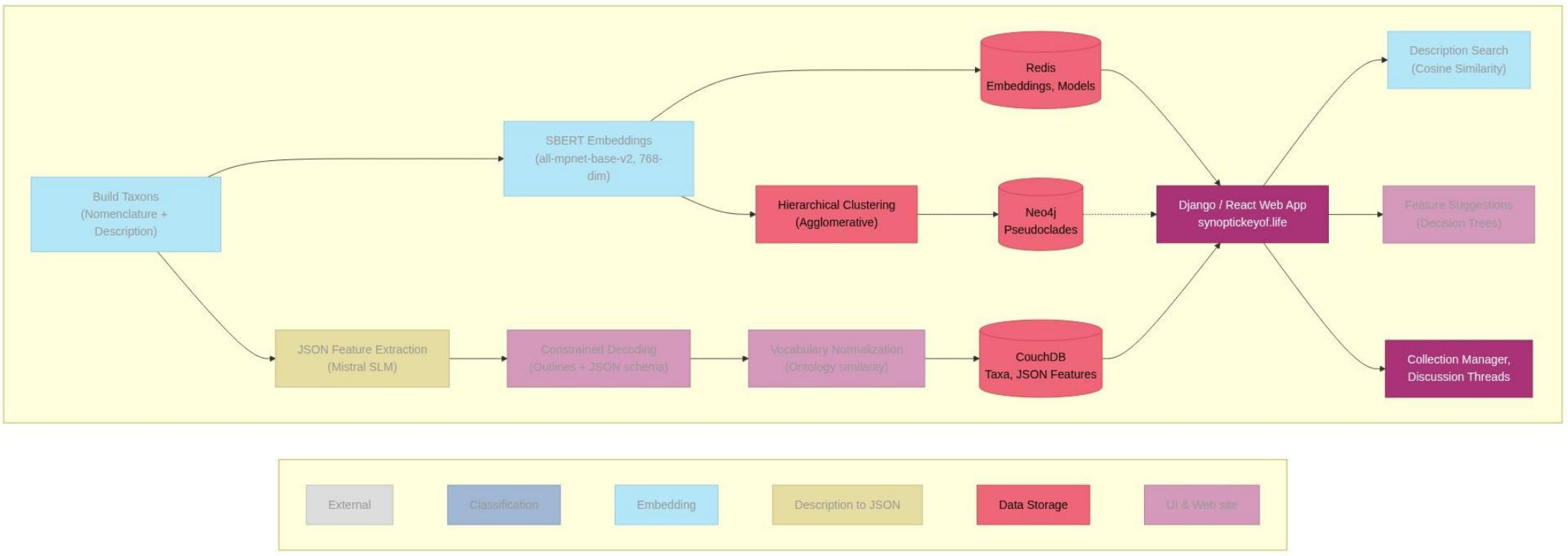
Description to JSON

Can you extract any attributes and their measurements from the following description of an organism? Present the result as a json data structure.
Mycelium on the substrate is medium orange-brown, septate, 3-4 µm diam., often distinctly and coarsely rough; hyphae often run parallel to the long axis of cells in substrate tissue but usually branch on the exterior and appear as a non-patterned network. Sporulation units are produced as enlarging tips of short branches, at first concolorous with surface mycelium but soon dark orange-brown and finally black-brown and opaque. In type material an occasional enlarging branch can be seen attached to the hypha from which it developed; in most instances, however, the enlarged branch elements have broken loose from the mycelium or are so densely crowded on the substrate and so nearly opaque that points of attachment cannot be traced


```
{
    "mycelium": {
        "location": "on substrate",
        "color": "medium orange-brown"
    },
    "hyphae": {
        "septate": true,
        "diameter_um": {
            "min": 3,
            "max": 4
        },
        "surface": "distinctly and coarsely rough",
        "orientation": "parallel to long axis of substrate tissue cells",
        "branching": {
            "location": "on exterior",
            "pattern": "non-patterned network"
        }
    },
    "sporulation_units": {
        "origin": "enlarging tips of short branches",
        "color_progression": [
            "concolorous with surface mycelium",
            "dark orange-brown",
            "black-brown and opaque"
        ],
        "attachment": {
            "occasional": "attached to hypha",
            "typical": "broken loose or obscured by crowding and opacity"
        }
    }
}
```


Description to JSON


Journal Web Sites (Mycotaxon, Persoonia, MycoKeys, IMA Fungus, Studies in Mycology, J. of Fungi, MycoWeb, Internet Archive)
Ingesters (RSS, Ingenta, Pensoft, MDPI, Internet Archive, CrossRef)
CouchDB Articles, PDFs, Metadata
PDF Text Extraction (PyMuPDF + OCR fallback)
PySpark Classifier (Logistic Regression, TF-IDF) Nomenclature | Description | Misc
Build Taxons (Nomenclature + Description)
External
Classification
Embedding
Description to JSON
Data Storage
UI & Web site


Build Taxons (Nomenclature + Description)
SBERT Embeddings (all-mpnet-base-v2, 768-dim)
Redis Embeddings, Models
Hierarchical Clustering (Agglomerative)
Neo4j Pseudoclades
Description Search (Cosine Similarity)
Django / React Web App synoptickeyof.life
Feature Suggestions (Decision Trees)
Collection Manager, Discussion Threads
JSON Feature Extraction (Mistral SLM)
Constrained Decoding (Outlines + JSON schema)
Vocabulary Normalization (Ontology similarity)
CouchDB Taxa, JSON Features
External
Classification
Embedding
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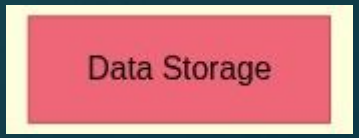
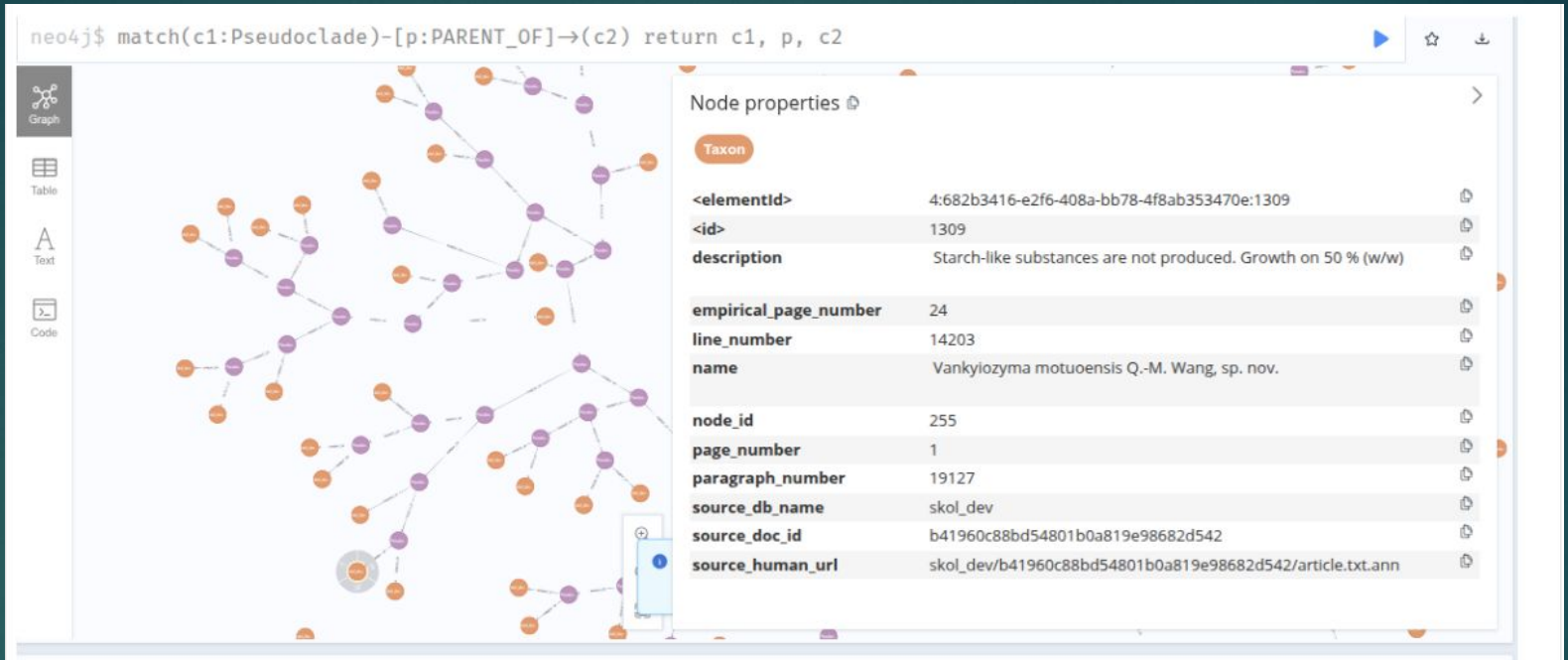
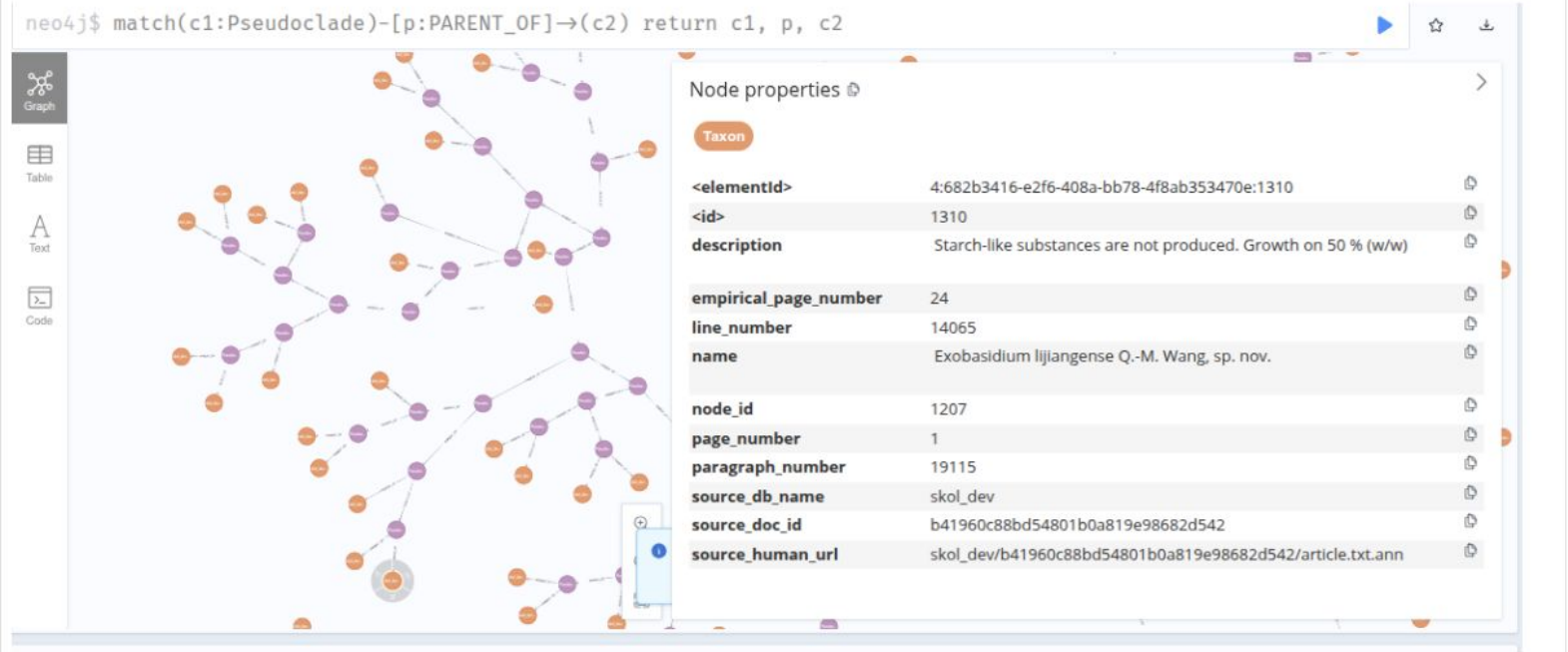

Data Storage

Pseudoclades





Data Storage

Search Page (showing JSON Feature Tool)

Welcome, **niemand** | My Collections | Help | Feedback | Contact | Account | Logout

SKOL Semantic Search α

Search taxonomic descriptions using semantic similarity

Try this:

```
campanulate pileus, stipe stains blue when cut
```

Active Collection: inat stereum × + New

ID: 601199613 View Details

Nomenclature: Unknown

Description Text (linked to collection) Flag

orange margin; hyphaesystem ketoatocalhyphae thickened walls; spores (7-) 10.5 - 11.3 (-12) × (6.5-) 9.5 - 10 (-11) µm, Q: (1-)1.1-1.1(1.1); paraphyses (100-) 116 - 137 (-147) µm;

Vocabulary | Text Features | JSON Features | Metrics

100 taxa, 739 features, depth 10

- [ ] conidia cylindrical shape 0 cylindrical
- [ ] growthform 11 description 0 epihymenium colourless, faintly yellowi
- [ ] meissner hydnoidea arrangement 0 opposite or alternate
- [ ] conidioma structure 1 1–4 mm diam.

UI & Web site

Search Results

Search | Leucocoprinus | List Names

▸ Discussion (0)

External Identifiers

GenBank 123456 Delete

Select type... | Identifier value (e.g., 336010515) | Add

Found 22 results

From collection search: "pileus form umbo low central umbo; pileus microstr..."

Loading 22 results from collection...

From: pileus form umbo low central umbo; pileus microstr...

Leucocoprinus tenellus Pegler [nom. illegit., non (Boud.) Locq.] Context 97.3%

Label Collection

Feed: User Collection

pileus form umbo low central umbo; pileus microstructure pileipellisform cutis;asci spores per ascus 4;spores (11-) 13 - 13 (-14) × (9.5-) 10 - 10 (-11) µm, Q: (1.1-)1.3-1.3(1.4); Context

Show Full JSON

Leucocoprinus tenellus Pegler [nom. illegit., non (Boud.) Locq.] from the Context 82.6%

Label Collection

Feed: CouchDB Taxa PDF Page: 7

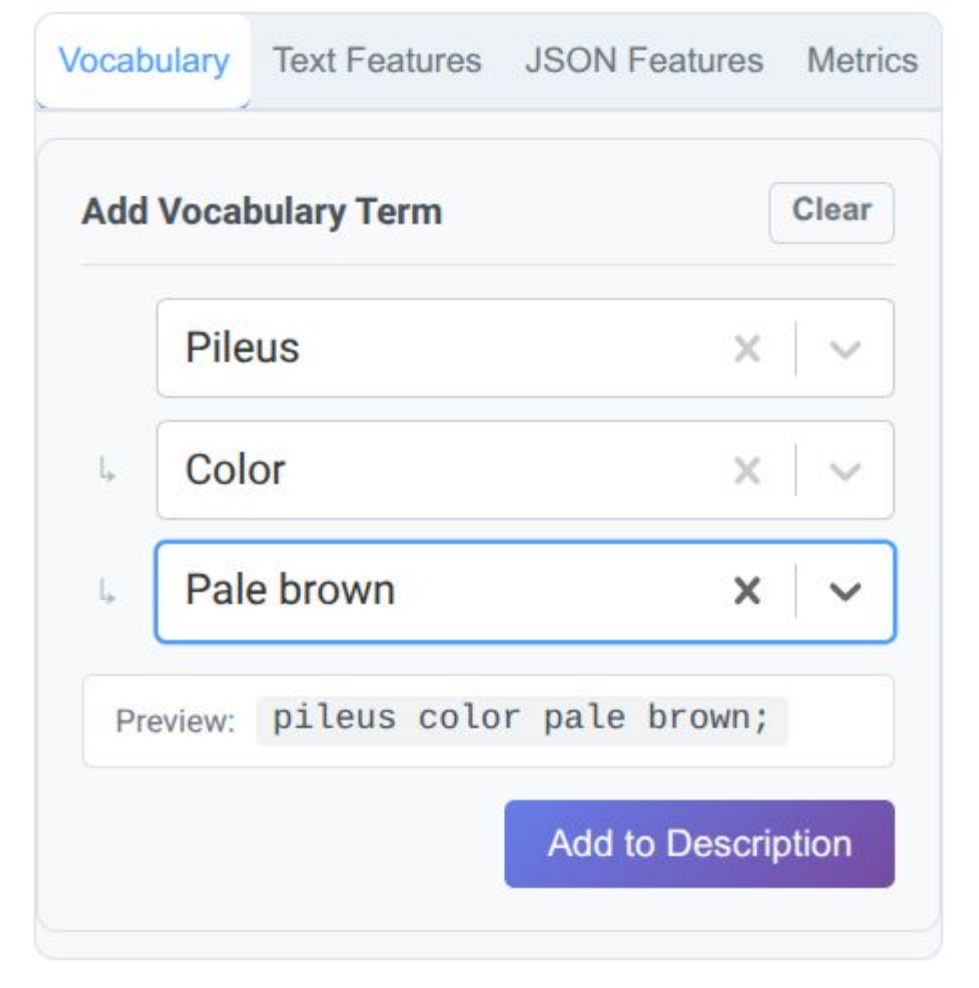
4b. Pileus white to yellowish-brown, pale brown or greyish; stipe whitish, Context
without reddish, greenish or bluish tinges; spores 5–8 µm in broad. 5
5a. Pileus 15–25 mm in diam., hemispheric to convex, pale brown with

stipe greyish white to yellowish, smooth or covered by

View PDF | View Source | Show Full JSON

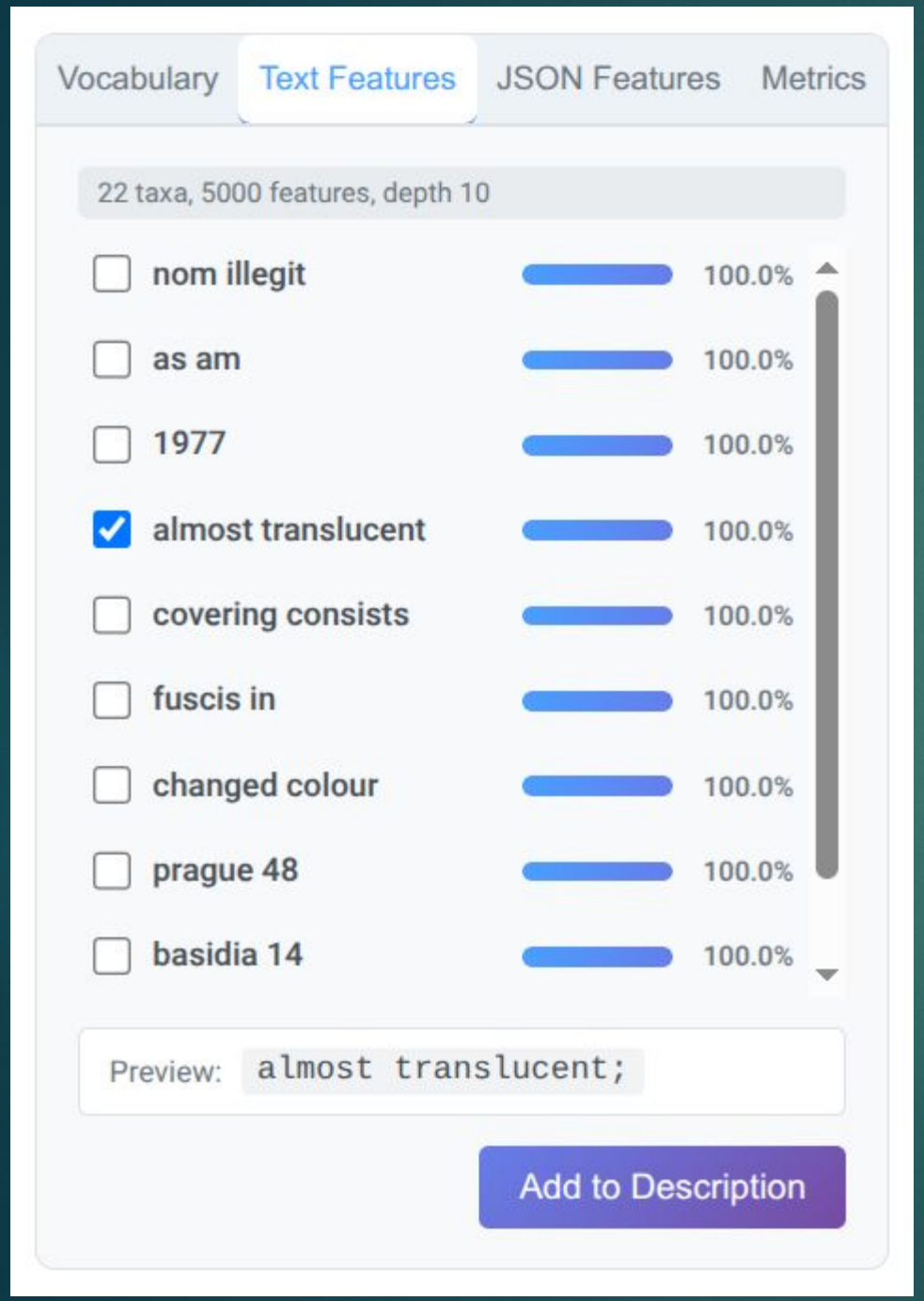
UI & Web site

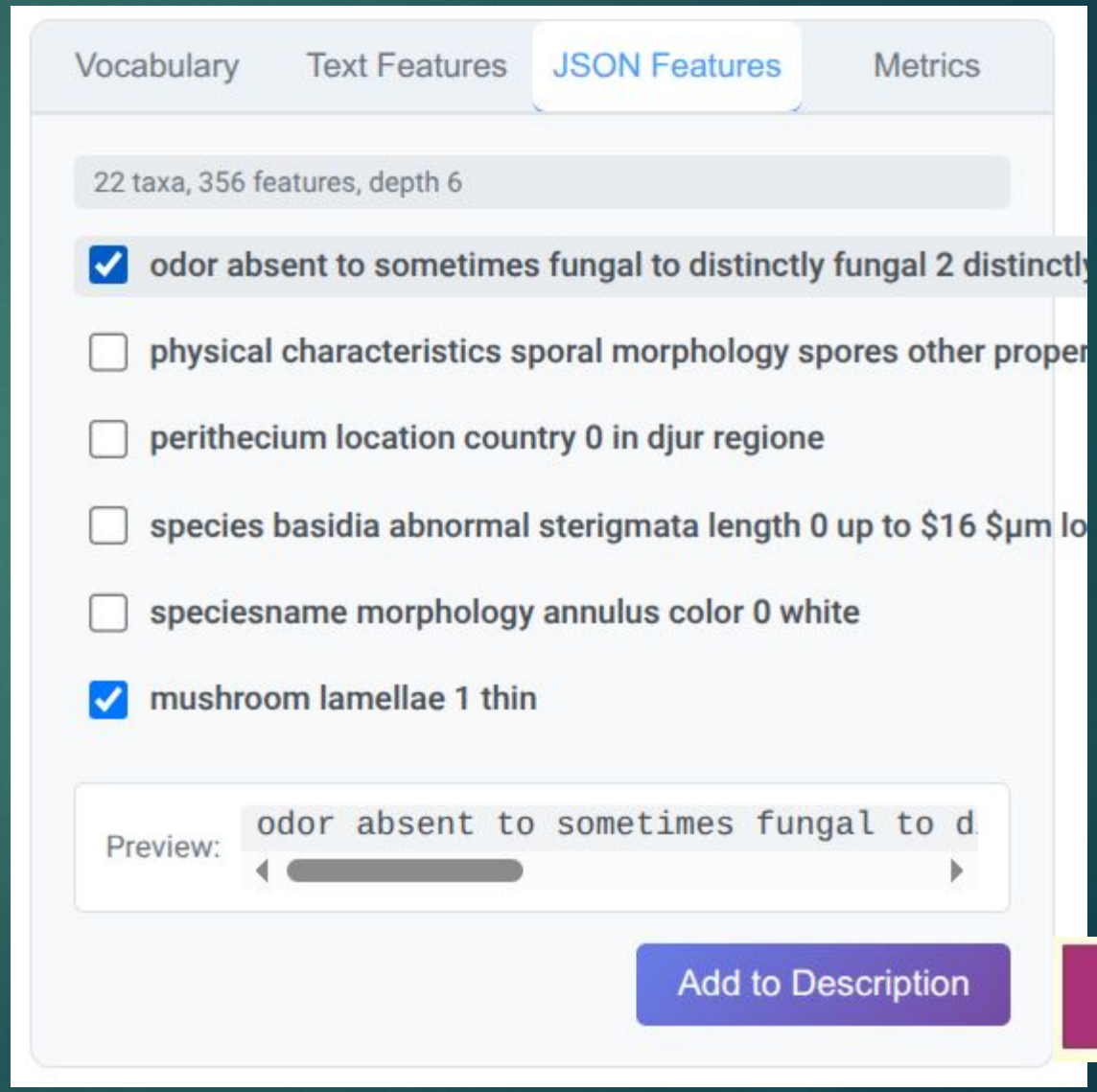
Vocabulary Feature Tool



UI & Web site

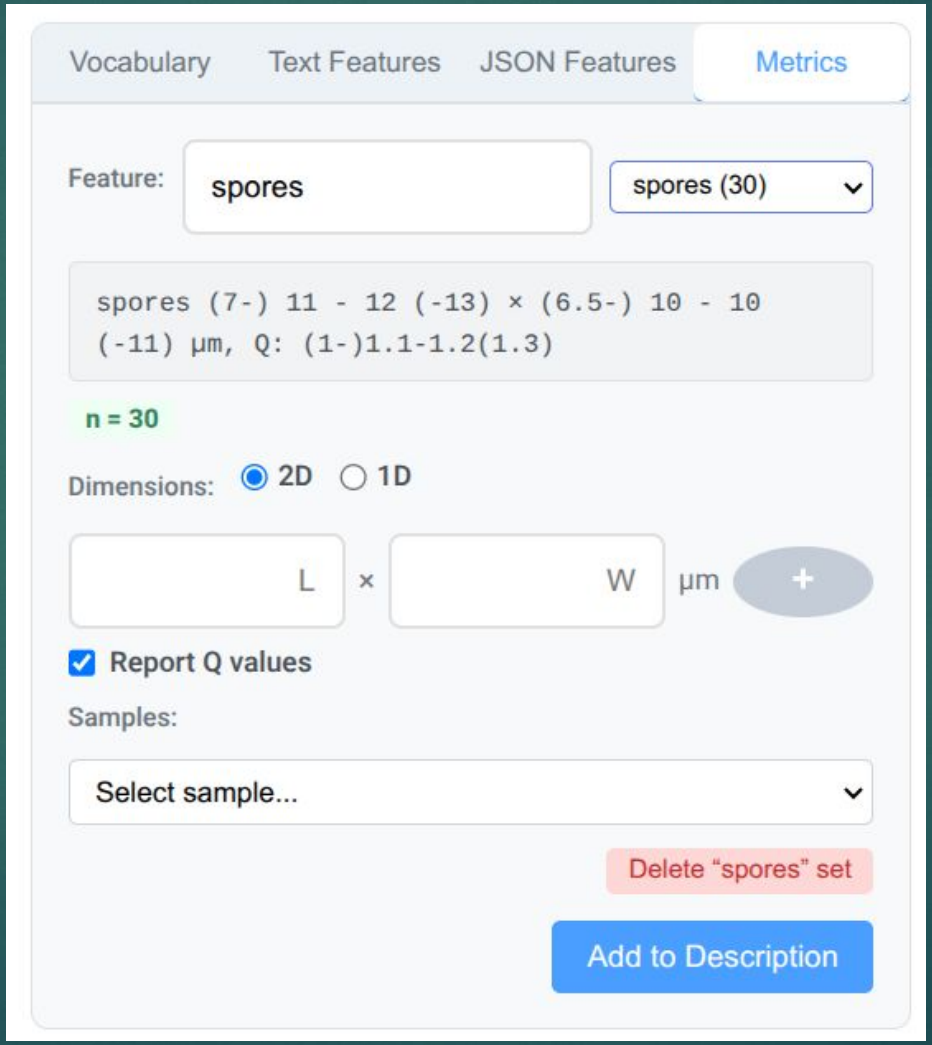
Text and JSON Feature tools





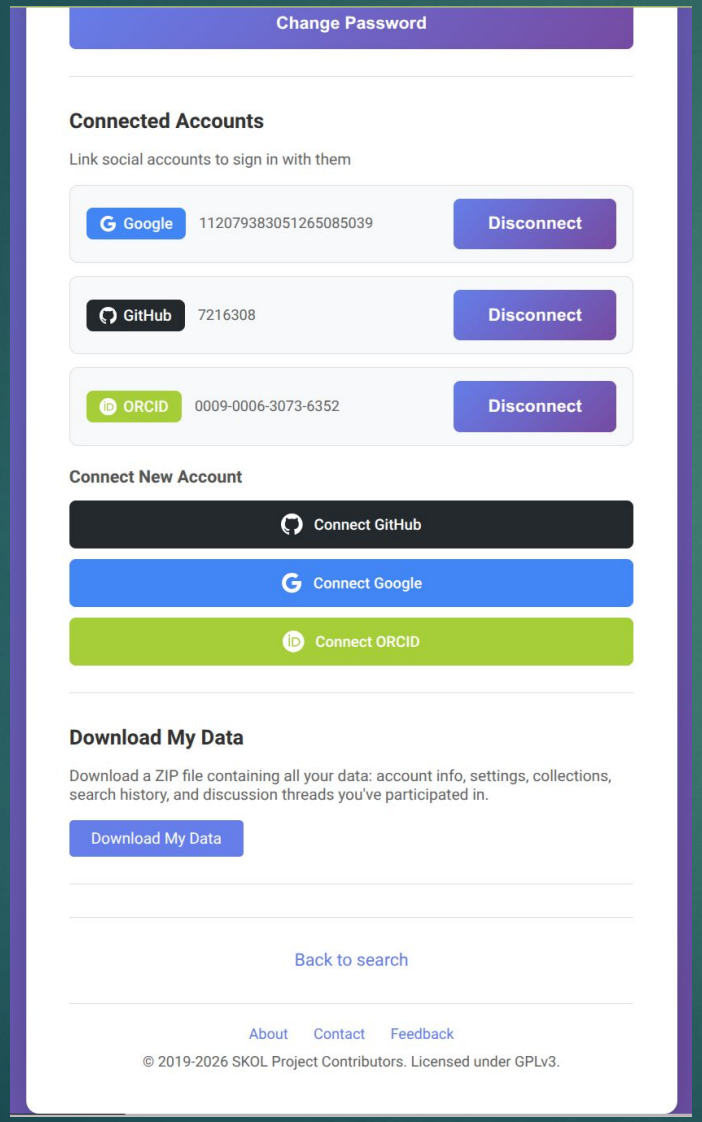
UI & Web site

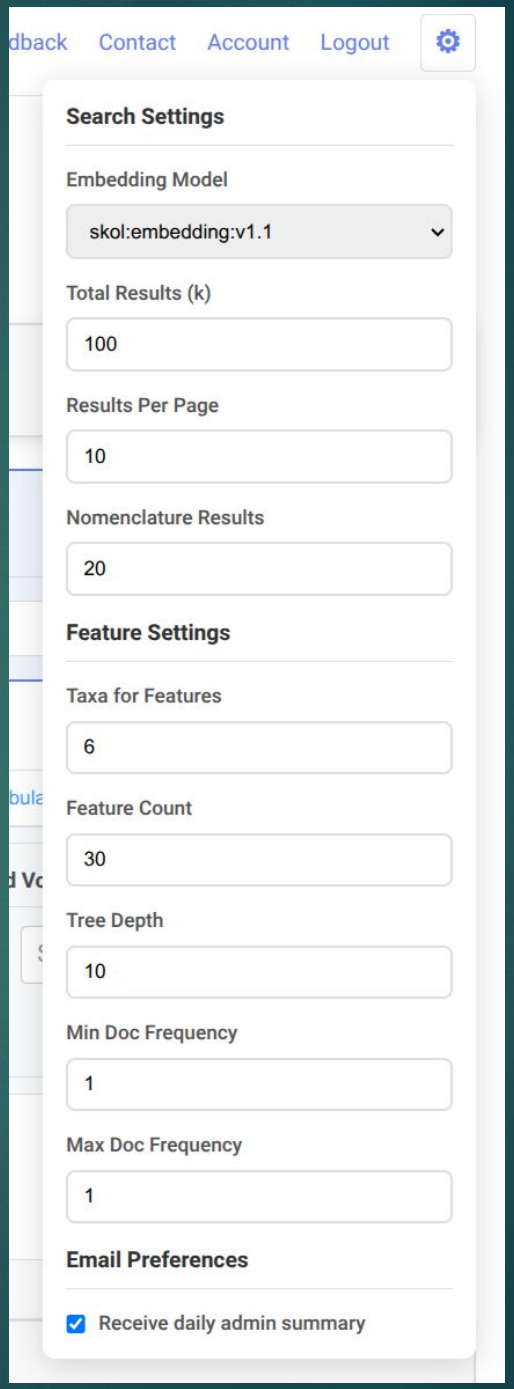
Metrics Description Tool



UI & Web site

Account and Settings





UI & Web site

Future

- Currently in alpha-test
- Data quality project, especially for older material
- Glossary tools
- Better multi-language support
- More journals and books
- Expand to other groups of organisms
- See https://trello.com/b/jUkRE6zv/skol for a complete backlog.

Acknowlegements

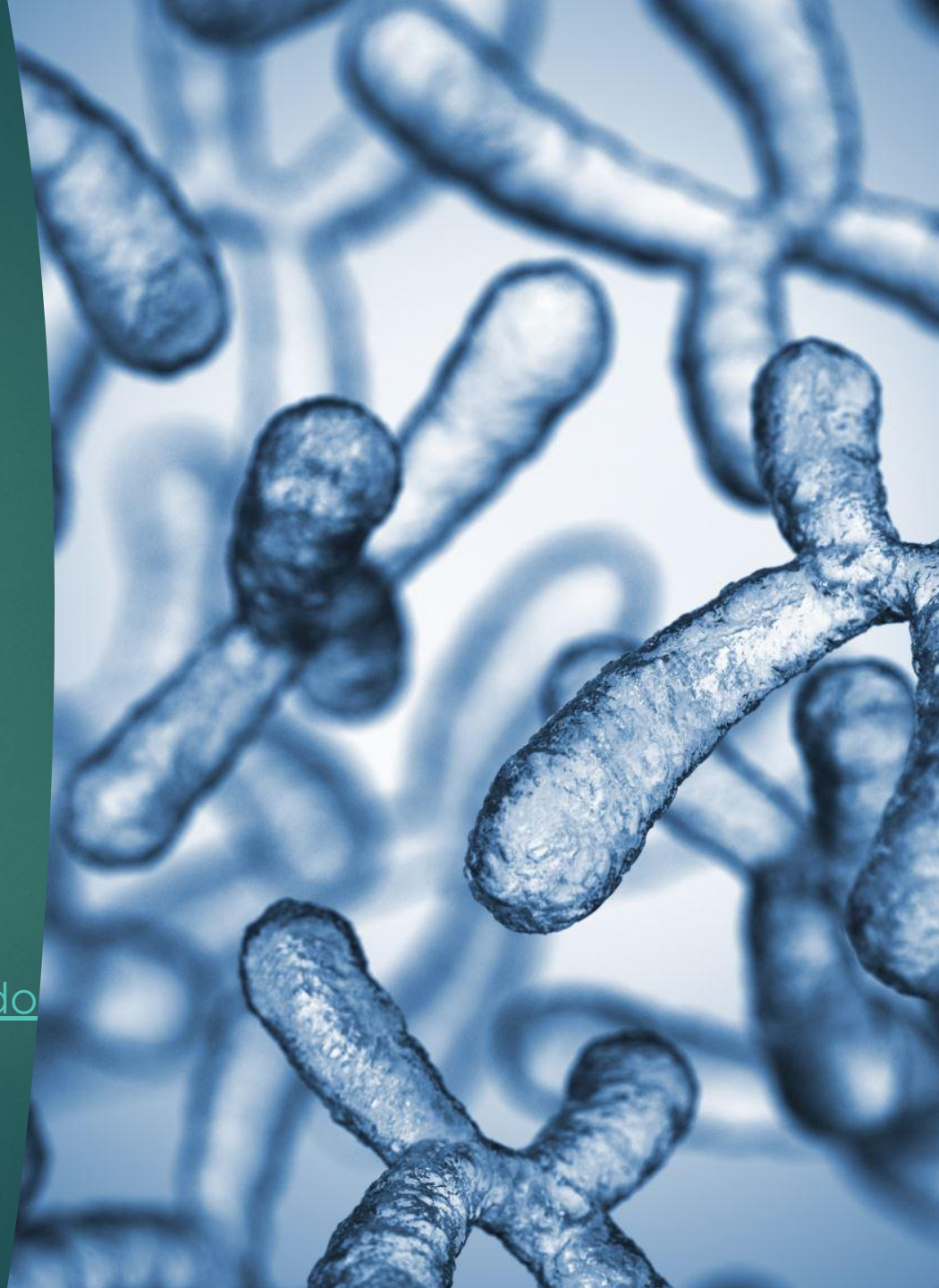
Development team

- Dr. Gregory Block (Faculty advisor)
- Christopher Murphy
- Jennifer Balasi
- Shintaro Osuga
- David Caspers
- Padmaja Kurumaddali
- Patrick Le

Alpha testers

- Garrett Taylor
- Richard Jacob

These slides:
https://tinyurl.com/ykwt8dkv
The accompanying paper:
https://tinyurl.com/54paunua
The SKOL web site:
https://synoptickeyof.life/skol
The SKOL source code:
https://github.com/piggyatbaqaqi/skol
Dr Draft's state-of-the-art (SOTA) Literature Search
https://github.com/autonlab/dr-drafts-sota-literature-search
SKOL fork of Dr Draft's:
https://github.com/piggyatbaqaqi/dr-drafts-mycosearch
A complete summary of technologies integrated:
https://github.com/piggyatbaqaqi/skol/blob/main/do



Appendix: Sources

Publication	Publisher/Source	Website	Total Documents	Taxonomy Documents	Taxa Documents
Cryptogamie, Mycologie	crossref	https://sciencepress.mnhn.f...	987	0 (0.0%)	8
Fungal Systematics and Evolution	ingenta	https://api.ingentaconnect.c...	139	0 (0.0%)	298
Journal of Fungi	mdpi	https://www.mdpi.com/journ...	5039	0 (0.0%)	387
Mycologia (Taylor & Francis)	crossref	https://www.tandfonline.co...	5078	0 (0.0%)	2
Mycology: An International Journal on Fungal Biology (Taylor & Francis)	crossref	https://www.tandfonline.co...	508	0 (0.0%)	13
Mycology: An International Journal on Fungal Biology (Taylor & Francis)	Unknown	—	15	0 (0.0%)	0
Mycoscience: An open access journal of the Mycological Society of Japan	crossref	https://mycoscience.org	2027	0 (0.0%)	39
Mycotaxon	ingenta	https://api.ingentaconnect.c...	2199	1881 (85.5%)	2002
Open Access Journal of Mycology & Mycological Sciences	medwin-publishers	https://www.medwinpublish...	112	0 (0.0%)	3
Persoonia	ingenta	https://api.ingentaconnect.c...	161	0 (0.0%)	1035
Studies in Mycology	ingenta	https://api.ingentaconnect.c...	53	45 (84.9%)	731
Sydowia (Internet Archive)	internet-archive	https://archive.org/details/p...	12	0 (0.0%)	27
Unknown	Unknown	—	7541	0 (0.0%)	15575